Machine Learning Project Ideas

🧠 1. Data Analytics & Predictive Modelling

Project	Description	Techniques / Tools
Employee Attrition Prediction	Predict employees likely to leave using HR data.	Logistic Regression, Random Forest, SHAP Explainability
Sales Forecasting System	Forecast future sales using time series.	ARIMA, LSTM, Prophet
Customer Lifetime Value Prediction	Estimate long-term revenue from each customer.	Gradient Boosting, XGBoost, Regression
Market Basket Analysis	Find product purchase patterns.	Association Rule Mining, Apriori, FP-Growth
Predictive Maintenance for Machines	Predict when machines need servicing using sensor logs.	Random Forest, XGBoost, Time Series
Stock Market Volatility Prediction	Predict volatility index (VIX) using historical and sentiment data.	LSTM, ARIMA, Sentiment Analysis

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💬 2. Natural Language Processing (NLP)

Project	Description	Techniques / Tools
Automated Resume Screening	Rank and classify resumes based on job descriptions.	BERT, TF-IDF, Cosine Similarity
Customer Support Chatbot	AI assistant for FAQs or complaint redressal.	Transformer, LangChain, RAG
Text Summarization for Research Papers	Generate concise summaries of long documents.	Pegasus, BART, T5
Multilingual Sentiment Analysis	Analyze product reviews in multiple languages.	mBERT, XLM-R

Project	Description	Techniques / Tools
Legal Document Classification	Automatically categorize contracts, case laws, etc.	BERT, Logistic Regression
Meeting Transcript Analyzer	Extract action items and key points from recorded meetings.	Whisper (ASR) + GPT Models

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🧬 3. Healthcare & Biomedical Projects

Project	Description	Techniques / Tools
Diabetes / Heart Disease Prediction	Predict risk level from medical parameters.	Random Forest, Logistic Regression
Cancer Detection from Histopathology Images	Classify tumor images.	CNN, Transfer Learning
ECG Anomaly Detection	Detect cardiac arrhythmia from ECG signals.	CNN + LSTM
AI-based Medical Report Generator	Generate diagnosis summaries from patient data.	NLP + LLMs
Drug Recommendation System	Suggest optimal drug combinations for patients.	Collaborative Filtering, Graph Neural Networks
Sleep Disorder Detection using Wearable Data	Predict poor sleep from accelerometer/HR data.	RNN, Time Series Forecasting

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🚗 4. Computer Vision & Image Processing

Project	Description	Techniques / Tools
Traffic Sign Recognition System	Detect and classify road signs in real-time.	CNN, YOLOv8
Real-Time Helmet Detection for Riders	Detect motorbike riders not wearing helmets.	YOLOv5, OpenCV

Project	Description	Techniques / Tools
Attendance System using Face Recognition	Mark attendance via webcam using facial ID.	FaceNet, Dlib, OpenCV
Garbage Classification for Smart Cities	Classify waste type from camera input.	CNN, Transfer Learning
Automatic Number Plate Recognition (ANPR)	Detect and read vehicle number plates.	OCR + YOLO
Defect Detection in Manufacturing	Detect defects in products on conveyor belts.	CNN, Autoencoders

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🛒 5. E-Commerce, Finance & Business

Project	Description	Techniques / Tools
Dynamic Pricing System	Adjust product prices based on demand and competitor pricing.	Reinforcement Learning, Regression
Personalized Product Recommender	Suggest products using user browsing history.	Collaborative Filtering, Deep Learning
Fraudulent Transaction Detection	Identify abnormal or risky transactions.	Isolation Forest, XGBoost
Customer Segmentation using RFM Analysis	Group customers for targeted marketing.	K-Means, Hierarchical Clustering
Demand Forecasting for Retail Chains	Predict demand to manage stock efficiently.	Prophet, XGBoost, LSTM
Invoice OCR & Auto Accounting System	Extract and process text from invoices.	OCR, NLP, Document AI

🌾 6. Agriculture & Environmental AI

Project	Description	Techniques / Tools
Crop Yield Prediction	Predict yield based on soil, rainfall, and temperature.	Regression, XGBoost
Crop Disease Detection	Identify diseases using leaf images.	CNN, MobileNet
Irrigation Automation using ML	Predict water requirement from weather and soil data.	Regression, IoT Integration
Forest Fire Detection from Satellite Images	Predict wildfire regions using image segmentation.	CNN, U-Net
Rainfall and Drought Forecasting	Predict long-term climate trends.	LSTM, Prophet
Fertilizer Recommendation System	Suggest fertilizer mix based on soil nutrient data.	Decision Tree, Ensemble Learning

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🧩 7. Emerging & Cutting-Edge Topics

Project	Description	Techniques / Tools
Federated Learning for Healthcare	Train models across hospitals without data sharing.	TensorFlow Federated
Explainable AI for Decision Making	Add interpretability layers (LIME, SHAP).	Explainable ML
Fake Image / Deepfake Detection	Detect manipulated images or videos.	CNN, GAN Discriminator Models
AI-based Cybersecurity Intrusion Detection	Detect malicious network activity.	Autoencoder, Isolation Forest
Autonomous Drone Path Optimization	RL-based navigation and obstacle avoidance.	Reinforcement Learning
AI-Powered Recruitment Assistant	Match job descriptions with candidate profiles.	NLP, BERT, Matching Algorithms

Project	Description	Techniques / Tools
AI for Energy Optimization in Smart Buildings	Predict and minimize energy usage.	Reinforcement Learning, Time Series